Band-1 Hadamard Order 8884 – Miyamoto (Paley-II Seed H_2220 q=1109 + C_2222); IN SPK via Paley-II; Gap 2052 Still OPEN

H_2220 Paley-II PASS; C_2222 PASS; H_8884 Miyamoto PASS; H_8884 Paley-II PASS -> IN SPK; skip 8788/8836 honest; gap 2052 OPEN; conjecture OPEN; GATE-04 LOCKED

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Document class. Scientific computational / construction certificate (Band-1).
Program. Open-mathematics collapse list * Human+AI attack order * ZoraASI mission-control.
Parents. `2026-09-16_Band1_Hadamard_8644_Miyamoto`; `2026-09-16_Band1_Hadamard_8564_PaleyI`.
Lab. `/workspace/experiments/band1/hadamard_8884_miyamoto/`.
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Honesty stamp (non-negotiable).
This memo claims **finite constructions of orders 2220 and 8884 only (verified max|HHˆT-nI|=0), via Miyamoto with Paley-II seed H_2220 (q=1109) + C_2222 over GF(2221); 8884 ALSO via classical Paley-II q=4441 -> IN SPK.** Honest skips: **8788** (GF(2197)=13ˆ3 unimplemented; `paley_core_prime_power` k=1,2 only); **8836** (seed Paley-II q=1103==3 mod4 fails codebase rule). Classical SPK for 8884: Paley-I **FAIL**s ($q = 8883 = 3^3 \cdot 7 \cdot 47$ NOT prime); Paley-II **PASS** (q = 4441 prime ==1 mod4); Kronecker **FAIL**s (4×2221). Seed **2220** folds into SPK when Paley-II seed verifies; **8884** folds **only** via classical Paley-II (never Miyamoto alone). It **does not** claim the full Hadamard conjecture. GATE-04 collection remains **LOCKED**. No RH / P≠NP / YM / Frankl / NS-solved / AC-solved claims. Gap **2052** remains **OPEN**. `clay_closed = false`. Prior SPK **2160+2140+8564 / 1064+2128 / ...** retained; Miyamoto-only **8644/8516/8452/7988/...** excluded.

Abstract

Finite constructions of orders **2220** and **8884**.

- H_{2220} = Paley-II ($q = 1109$) – preferred Miyamoto seed
- C_{2222} = Paley conference over GF(2221), $q = 2221$
- H_{8884} = Miyamoto(H_{2220}, C_{2222})
- Paley-II for 8884: **PASS** ($q = 4441$ prime ==1 mod 4) – classical SPK
- Paley-I for 8884: **SKIP** ($q = 8883 = 3^3 \cdot 7 \cdot 47$ NOT prime)
- Kronecker for 8884: **SKIP** ($4 \cdot 2221$ not admissible)
- Skips: **8788** (GF(13^3)); **8836** (seed q=1103==3)

SPK: IN SPK via Paley-II.
Gap 2052: OPEN. **Conjecture:** OPEN. **clay_closed:** false. **GATE-04:** LOCKED.

`python3 verify_hadamard_8884.py`

Results. overall_pass=True; units 165/165; wall 1175.4792s; all constructed max|HH^T-nI|=0.

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Arithmetic

- Target $n = 8884 = 4 \cdot 2221$; 2221 prime $\equiv 1 \pmod 4$ -> Miyamoto-eligible.
- Need H_{2220} and C_{2222} (Paley conference over prime field $q = 2221$).
- Seed: Paley-II $q = 1109$ (preferred); Paley-I $q = 2219$ FAIL; Double FAIL; Kronecker 4×555 FAIL.
- SPK: Paley-I $q = 8883$ **FAIL**; Paley-II $q = 4441$ **PASS**; Kronecker **FAIL**.
- Prior optional skipped: 8788 (GF(13^3)); 8836 (seed ==3 mod4).

Constructions (verified)

Object	Method	max|HH^T-nI|	Wall (s)
H_2220 Paley-II	Paley-II q=1109	0	6.1027
C_2222	Paley conf GF(2221)	0	7.6178
H_8884 Miyamoto	Miyamoto(H_2220, C_2222)	0	567.7285
H_8884 Paley-II	Paley-II q=4441 classical SPK	0	579.9847
H_8884 Paley-I	SKIP q=8883=3^3*7*47	–	–
H_8884 Kronecker	SKIP 4*2221 not admissible	–	–

SPK closure note

Seeds include prior SPK through **8644** lineage (**2160**; **2140+8564**; **1064+2128**; ...; **NOT** 8644/8516/8452/...) + verified **2220**; fold **8884** only if Paley-II verifies. Next constructive gap **2052** (OPEN).

Quantity	Value
8884 in SPK	**True** (Paley-II q=4441)
2220 in SPK	**True** (Paley-II seed)
Gap 2052	**OPEN**
clay_closed	false
GATE-04	LOCKED

Honesty

- clay_closed = false
- GATE-04 LOCKED
- Hadamard conjecture OPEN
- Gap 2052 OPEN (method triage / non-finding only)
- No RH / P!=NP / YM / Frankl / NS / AC solved claims
- SPK membership only via classical Paley-I/II/Kronecker (not Miyamoto alone)

Next optional

q=2237->n=8948 PaleyII(q=1117) OR q=2269->n=9076 PaleyI(q=2267) OR q=2273->n=9092 Kronecker(4x568) OR q=2281->n=9124 triage OR method-attack SPK gap 2052 OR Track B / Object 1 meter await